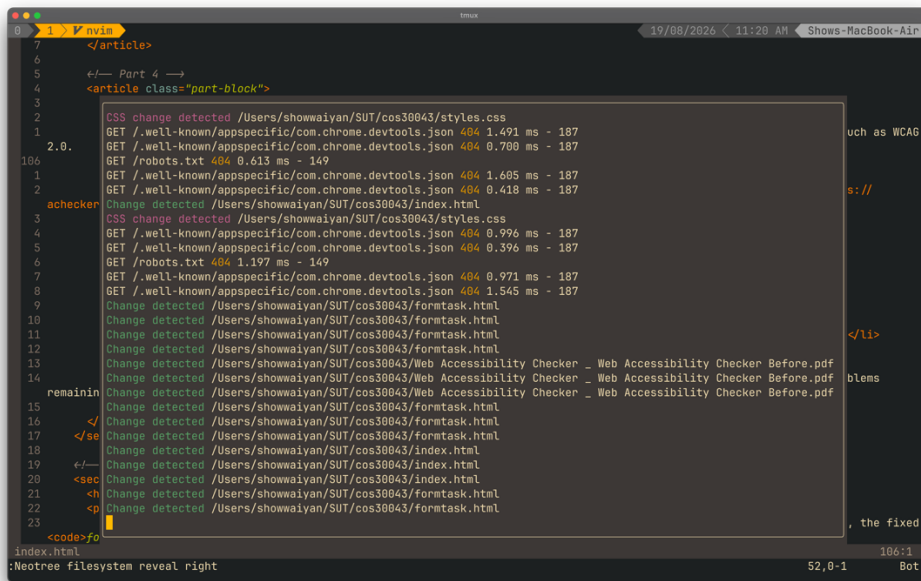


COS30043 Interface Design and Development

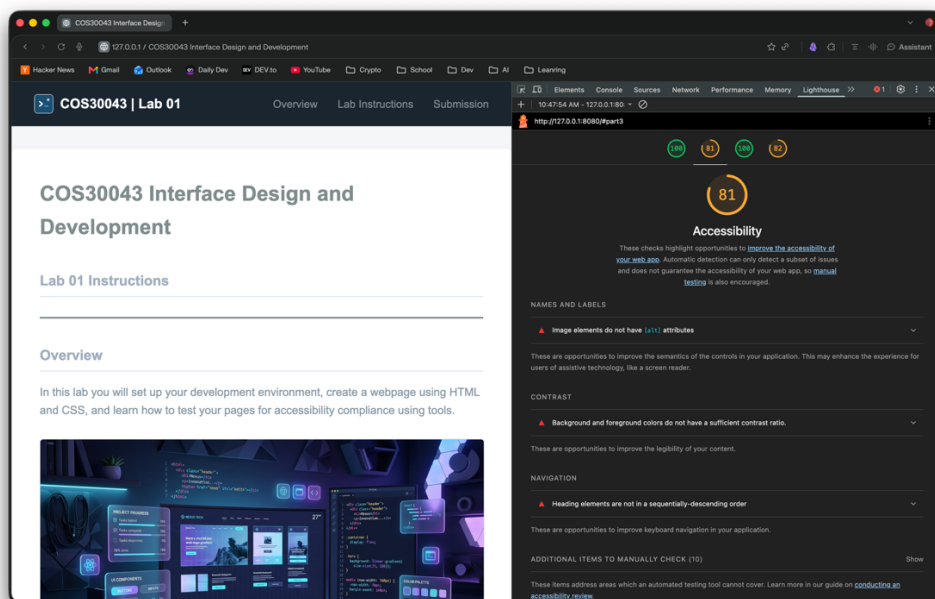
Lab 01

Liver Server Running

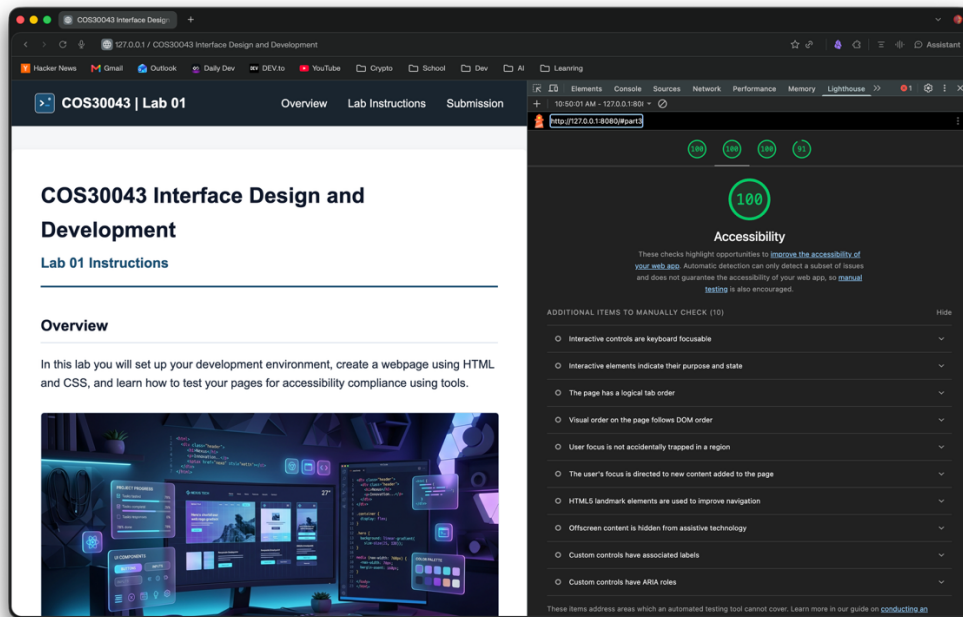


```
0 1> nvim
7 </article>
6
5 <!-- Part 4 -->
4 <article class="part-block">
3
2
1 CSS change detected /Users/showwaiyan/SUT/cos30043/styles.css
2.0. GET /.well-known/appspecific/com.chrome.devtools.json 404 1.491 ms - 187
186 GET /.well-known/appspecific/com.chrome.devtools.json 404 0.780 ms - 187
1 GET /robots.txt 404 0.613 ms - 149
2 GET /.well-known/appspecific/com.chrome.devtools.json 404 1.685 ms - 187
186 GET /.well-known/appspecific/com.chrome.devtools.json 404 0.418 ms - 187
1 Change detected /Users/showwaiyan/SUT/cos30043/index.html
3 CSS change detected /Users/showwaiyan/SUT/cos30043/styles.css
14 Change detected /Users/showwaiyan/SUT/cos30043/index.html
186 GET /.well-known/appspecific/com.chrome.devtools.json 404 0.996 ms - 187
1 GET /robots.txt 404 1.197 ms - 149
2 GET /.well-known/appspecific/com.chrome.devtools.json 404 0.971 ms - 187
186 GET /.well-known/appspecific/com.chrome.devtools.json 404 1.545 ms - 187
1 Change detected /Users/showwaiyan/SUT/cos30043/formtask.html
10 Change detected /Users/showwaiyan/SUT/cos30043/formtask.html
11 Change detected /Users/showwaiyan/SUT/cos30043/formtask.html
12 Change detected /Users/showwaiyan/SUT/cos30043/formtask.html
13 Change detected /Users/showwaiyan/SUT/cos30043/Web Accessibility Checker - Web Accessibility Checker Before.pdf
14 Change detected /Users/showwaiyan/SUT/cos30043/Web Accessibility Checker - Web Accessibility Checker Before.pdf
15 Change detected /Users/showwaiyan/SUT/cos30043/formtask.html
16 Change detected /Users/showwaiyan/SUT/cos30043/formtask.html
17 Change detected /Users/showwaiyan/SUT/cos30043/formtask.html
18 Change detected /Users/showwaiyan/SUT/cos30043/index.html
19 Change detected /Users/showwaiyan/SUT/cos30043/index.html
20 Change detected /Users/showwaiyan/SUT/cos30043/index.html
21 Change detected /Users/showwaiyan/SUT/cos30043/formtask.html
22 Change detected /Users/showwaiyan/SUT/cos30043/formtask.html
23
<code>fo
index.html
Neotree filesystem reveal right
52,0-1
Bot
```

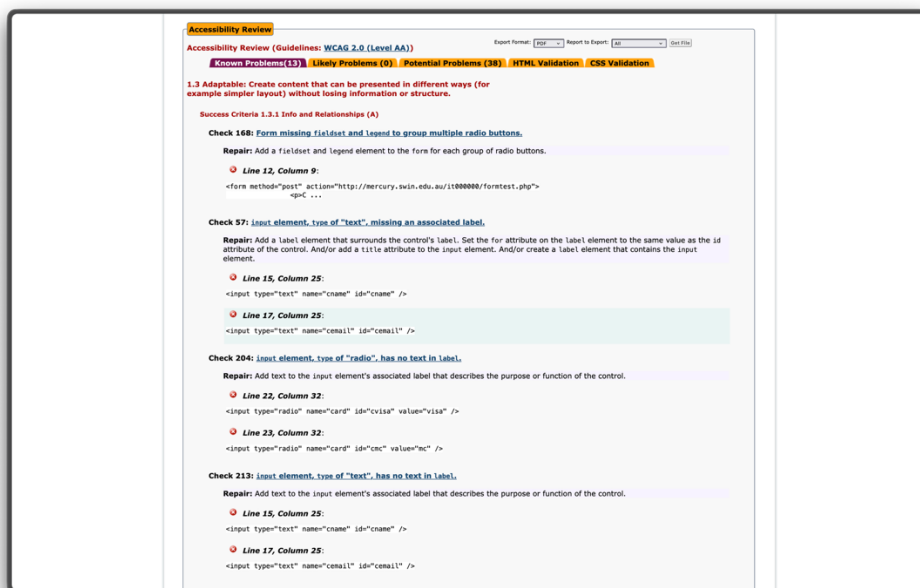
Lighthouse Score Before

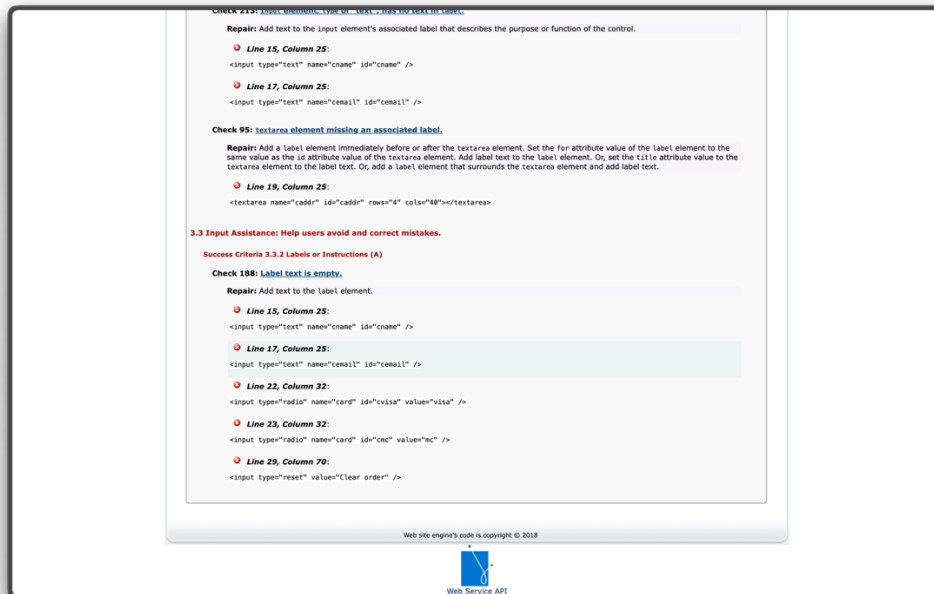


Lighthouse Score After



A Checker Issues Before for formtask.html





```

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>COS30043 Interface Design and Development - Lab 01</title>

<link rel="stylesheet" href="styles.css">

</head>

<body>

  <!-- Navigation Bar with Logo and Max 3 Links -->

  <header class="navbar">

    <div class="logo">

      <span class="logo-text">COS30043 | Lab 01</span>

    </div>

    <nav aria-label="Main Navigation">

      <a href="#overview">Overview</a>

      <a href="#instructions">Lab Instructions</a>

      <a href="#submission">Submission</a>

    </nav>

  </header>

  <!-- Page Title and Main Content Body -->

  <main>

    <header class="doc-header">

      <h1>COS30043 Interface Design and Development</h1>

      <h2>Lab 01 Instructions</h2>

    </header>

    <!-- Overview Section -->

    <section id="overview" aria-labelledby="overview-heading">

      <h3 id="overview-heading">Overview</h3>

      <p>

        In this lab you will set up your development environment, create a webpage using HTML and CSS, and learn how to test your pages
        for accessibility compliance using tools.

      </p>

      <figure>

        <!-- Added descriptive alt attribute to fix image accessibility -->

        <figcaption>Figure 1: Development workspace setup with live server preview</figcaption>

      </figure>

    </section>

    <!-- Lab Instructions Section (Parts 1 to 4) -->

    <section id="instructions" aria-labelledby="instructions-heading">

      <h3 id="instructions-heading">Lab Instructions & Tasks</h3>

      <!-- Part 1: Heading h4 sequentially follows h3 -->

```

```

<article class="part-block">

  <h4>Part 1: Setting Up Your Development Environment</h4>

  <p>

    If you have not already done so, install Visual Studio Code from

    <a href="https://code.visualstudio.com/" target="_blank" rel="noopener noreferrer">https://code.visualstudio.com/</a>.

  </p>

  <p>Install the following recommended extensions:</p>

  <ul>

    <li>

      <strong>Live Server:</strong> Live Server can launch a local development server with live reload. If you right-click an HTML
      file and select &ldquo;Open with Live Server,&rdquo; you can preview your page in the browser.

    </li>

  </ul>

</article>

<!-- Part 2 -->

<article class="part-block">

  <h4>Part 2: Create a Webpage</h4>

  <p>

    Create a single webpage using HTML and CSS to refresh what you learned before. You may also incorporate other techniques you
    know (e.g. JavaScript), but HTML and CSS are the minimum requirement.

  </p>

  <p>Your page must include the following elements:</p>

  <ul>

    <li>A logo.</li>

    <li>A page title.</li>

    <li>A navigation bar with links.</li>

    <li>A footer with relevant information (e.g. copyright, contact details).</li>

    <li>The main body of the page must include at least: a paragraph, an image, a table, and a list.</li>

    <li>Use appropriate styles.</li>

  </ul>

  <p>

    Preview your page using VS Code Live Server. Continue refining it until you are satisfied with the layout and content.

  </p>

</article>

<!-- Part 3 -->

<article class="part-block">

  <h4>Part 3: Accessibility Testing with Lighthouse</h4>

  <p>

    Lighthouse is an open-source auditing tool built into Google Chrome&rsquo;s DevTools. It evaluates web pages for performance,
    accessibility, best practices, and SEO, and provides actionable recommendations for improvement.

  </p>

  <ol>

    <li>Open your webpage in Google Chrome using Live Server.</li>

    <li>Open Chrome DevTools (right-click the page and select &ldquo;Inspect&rdquo;).</li>

    <li>Navigate to the &ldquo;Lighthouse&rdquo; tab in DevTools.</li>

  </ol>

```

```
    <li>Under &ldquo;Categories&rdquo;;, ensure &ldquo;Accessibility&rdquo; is selected (you may deselect others to speed up the audit).</li>
```

```
    <li>Click &ldquo;Analyze page load&rdquo; to generate the report.</li>
```

```
    <li>Review the accessibility score and the list of issues found.</li>
```

```
    <li>Address the issues in your HTML/CSS, save, and re-run the audit until the score improves.</li>
```

```
</ol>
```

```
</article>
```

```
<!-- Part 4 -->
```

```
<article class="part-block">
```

```
    <h4>Part 4: Accessibility Testing with AChecker</h4>
```

```
    <p>
```

```
        AChecker is a web-based accessibility evaluation tool that checks HTML content against accessibility standards such as WCAG 2.0.
```

```
    </p>
```

```
    <ol>
```

```
        <li>Go to <a href="https://achecker.achecks.ca/checker/index.php" target="_blank" rel="noopener noreferer">https://achecker.achecks.ca/checker/index.php</a>.</li>
```

```
        <li>Select &ldquo;HTML File Upload&rdquo; and choose the attached &ldquo;formtask.html&rdquo;.</li>
```

```
        <li>Expand the &ldquo;Options&rdquo; section and select &ldquo;WCAG 2.0 (Level AA)&rdquo;.</li>
```

```
        <li>Click &ldquo;Check It&rdquo;.</li>
```

```
    </ol>
```

```
    <p>The Accessibility Review will categorise issues into three types:</p>
```

```
    <ul>
```

```
        <li><strong>Known Problems:</strong> Definite accessibility barriers that must be fixed.</li>
```

```
        <li><strong>Likely Problems:</strong> Issues that are very probably barriers and should be reviewed.</li>
```

```
        <li><strong>Potential Problems:</strong> Items that may be barriers depending on context; review is recommended.</li>
```

```
    </ul>
```

```
    <p>
```

```
        Review the Known Problems and fix them. Re-test your page after each round of fixes until there are no Known Problems remaining.
```

```
    </p>
```

```
</article>
```

```
</section>
```

```
<!-- Submission & Assessment Section -->
```

```
<section id="submission" aria-labelledby="submission-heading">
```

```
    <h3 id="submission-heading">Submission & Assessment</h3>
```

```
    <p>
```

```
        Submit your work to Canvas, including your website source code, a screenshot of the Lighthouse accessibility score, the fixed <code>formtask.html</code>, and an AChecker report.
```

```
    </p>
```

```
    <p>
```

```
        Show your work to your tutor during the lab session to get marked off. This lab is assessed in Week 1 and Week 2 only. If you cannot attend, please provide supporting documentation (e.g. a medical certificate) to request an extension.
```

```
    </p>
```

```
    <p>
```

```
        Your tutor will review your work and ask you questions to assess your understanding. Marks are based on the quality of your work and your ability to explain your work.
```

```
</p>

<!-- Assessment Rubric Table -->

<table>

  <caption>COS30043 Lab 01 Assessment Rubric</caption>

  <thead>

    <tr>

      <th scope="col">Marks</th>

      <th scope="col">Criteria & Demonstration of Understanding</th>

    </tr>

  </thead>

  <tbody>

    <tr>

      <td>Up to 2 marks</td>

      <td>Limited or no understanding demonstrated.</td>

    </tr>

    <tr>

      <td>Up to 4 marks</td>

      <td>Reasonable understanding demonstrated.</td>

    </tr>

    <tr>

      <td>Up to 6 marks</td>

      <td>Strong and clear understanding demonstrated.</td>

    </tr>

  </tbody>

</table>

</section>

</main>

<!-- Footer with Relevant Information -->

<footer>

  <p>Contact: <a href="mailto:student@swinburne.edu.au">student@swinburne.edu.au</a> | Swinburne University of Technology</p>

  <p>&copy; 2026 COS30043 Interface Design and Development. All rights reserved.</p>

</footer>

</body>

</html>
```

Style.css

```
/* Fully Accessible & High-Contrast (WCAG AAA / Lighthouse 100/100) CSS for COS30043 Lab 01 */

body {

  font-family: Arial, sans-serif;

  line-height: 1.6;

  margin: 0;

  padding: 0;
```

```
    color: #1e293b; /* 12.8:1 AAA contrast ratio against #f4f6f8 */

    background-color: #f4f6f8;
}
```

```
/* Focus States for Keyboard Navigation */
```

```
a:focus-visible {

    outline: 3px solid #1b4f72;

    outline-offset: 3px;

    border-radius: 2px;
}
```

```
/* Top Navigation Bar (Non-sticky, exactly 3 links) */
```

```
.navbar {

    background-color: #1a252f; /* Deep dark slate background */

    color: #ffffff;

    padding: 1rem 2rem;

    display: flex;

    justify-content: space-between;

    align-items: center;
}
```

```
.logo {

    display: flex;

    align-items: center;

    gap: 0.6rem;

    font-size: 1.2rem;

    font-weight: bold;

    color: #ffffff;
}
```

```
.logo-img {

    width: 32px;

    height: 32px;

    display: block;
}
```

```
nav a {

    color: #ffffff; /* 15.6:1 contrast ratio against #1a252f */

    text-decoration: none;

    margin-left: 1.5rem;

    font-weight: 500;

    font-size: 0.95rem;
}
```

```
nav a:hover, nav a:focus {  
    text-decoration: underline;  
    color: #a9cce3; /* 8.6:1 AAA contrast ratio */  
}
```

```
/* Main Content Container */
```

```
main {  
    max-width: 820px;  
    margin: 2rem auto;  
    padding: 2.5rem;  
    background-color: #ffffff;  
    box-shadow: 0 2px 10px rgba(0,0,0,0.06);  
    border-radius: 8px;  
}
```

```
/* Document Header (Page Title & Subtitle) */
```

```
.doc-header {  
    border-bottom: 3px solid #1b4f72;  
    padding-bottom: 1rem;  
    margin-bottom: 2rem;  
    background: transparent;  
    display: block;  
}
```

```
.doc-header h1 {  
    color: #0f172a; /* 18.2:1 AAA contrast ratio */  
    font-size: 1.85rem;  
    font-weight: 700;  
    margin: 0 0 0.4rem 0;  
    border-bottom: none;  
    padding: 0;  
}
```

```
.doc-header h2 {  
    color: #1b4f72; /* 7.2:1 AAA contrast ratio */  
    font-size: 1.2rem;  
    font-weight: 600;  
    margin: 0;  
}
```

```
/* Sequential Section Headings (h3) */
```

```
h3 {
```

```
color: #0f172a; /* 18.2:1 AAA contrast ratio */

font-size: 1.3rem;

margin-top: 2.25rem;

margin-bottom: 0.75rem;

padding-bottom: 0.35rem;

border-bottom: 2px solid #e2e8f0;
}
```

```
.part-block {

margin-top: 1.25rem;

margin-bottom: 1.5rem;

padding-bottom: 1rem;

border-bottom: 1px dashed #cbd5e1;
}
```

```
.part-block:last-child {

border-bottom: none;
}
```

```
/* Sequential Subsection Headings (h4) */

h4 {

color: #1b4f72; /* 7.2:1 AAA contrast ratio */

font-size: 1.1rem;

font-weight: 700;

margin-top: 1rem;

margin-bottom: 0.5rem;
}
```

```
p {

margin-bottom: 1rem;

color: #1e293b; /* 12.8:1 AAA contrast ratio */
}
```

```
a {

color: #1b4f72; /* 7.2:1 AAA contrast ratio */

text-decoration: underline;

font-weight: 600;
}
```

```
a:hover, a:focus {

color: #0b2545; /* 14.5:1 AAA contrast ratio */
}
```

```
code {  
  
  font-family: 'Courier New', Courier, monospace;  
  
  background-color: #f1f5f9;  
  
  color: #821c14; /* 8.5:1 AAA contrast ratio */  
  
  padding: 0.15rem 0.4rem;  
  
  border-radius: 4px;  
  
  border: 1px solid #cbd5e1;  
  
  font-size: 0.9em;  
  
  font-weight: 600;  
  
}
```

```
figure {  
  
  margin: 1.75rem 0;  
  
  text-align: center;  
  
}
```

```
img {  
  
  max-width: 100%;  
  
  height: auto;  
  
  border-radius: 6px;  
  
  border: 1px solid #cbd5e1;  
  
}
```

```
figcaption {  
  
  font-size: 0.85rem;  
  
  color: #334155; /* 9.5:1 AAA contrast ratio */  
  
  margin-top: 0.5rem;  
  
  font-weight: 500;  
  
}
```

```
ul, ol {  
  
  padding-left: 1.5rem;  
  
  margin-bottom: 1.25rem;  
  
}
```

```
li {  
  
  margin-bottom: 0.5rem;  
  
  color: #1e293b; /* 12.8:1 AAA contrast ratio */  
  
}
```

```
table {  
  
  width: 100%;  
  
  border-collapse: collapse;
```

```

    margin-top: 1.25rem;

    margin-bottom: 1.5rem;
}

caption {

    font-weight: bold;

    margin-bottom: 0.5rem;

    text-align: left;

    color: #0f172a; /* 18.2:1 AAA contrast ratio */
}

th, td {

    border: 1px solid #cbd5e1;

    padding: 0.75rem 1rem;

    text-align: left;
}

th {

    background-color: #1a252f;

    color: #ffffff; /* 15.6:1 AAA contrast ratio */

    font-weight: bold;
}

td {

    color: #1e293b; /* 12.8:1 AAA contrast ratio */
}

tr:nth-child(even) {

    background-color: #f8fafc;
}

footer {

    text-align: center;

    padding: 2rem 1rem;

    background-color: #1a252f;

    color: #ffffff;

    margin-top: 3rem;

    font-size: 0.9rem;
}

footer p {

    color: #ffffff; /* 15.6:1 AAA contrast ratio */

    margin: 0.3rem 0;
}

```

```

}

footer a {
  color: #a9cce3; /* 8.6:1 AAA contrast ratio */
  font-weight: bold;
}

footer a:hover, footer a:focus {
  color: #ffffff;
}

@media (max-width: 650px) {
  .navbar {
    flex-direction: column;
    gap: 0.75rem;
    text-align: center;
    padding: 1rem;
  }

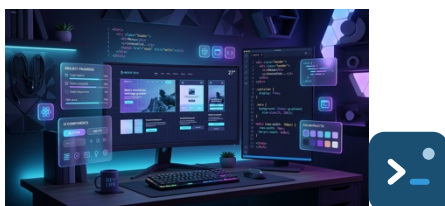
  nav a {
    margin: 0 0.5rem;
    font-size: 0.9rem;
  }

  main {
    margin: 1rem;
    padding: 1.25rem;
  }

  .doc-header h1 {
    font-size: 1.4rem;
  }
}

```

Images and Logo



formtask.html

```
<!DOCTYPE html>

<html lang="en" dir="ltr">

<head>

    <title>WAI Form Task - COS30043 Accessible Form</title>

    <meta charset="utf-8" />

    <meta name="language" content="english" />

    <meta name="keywords" content="Accessibility,Lab,Tasks,WAI,Form" />

    <meta name="description" content="WAI Accessible Order Form Task for COS30043 Interface Design and Development" />

    <style>

        /* Skip-to-content link – visible on focus only (WCAG 2.4.1) */

        .skip-link {

            position: absolute;

            top: -40px;

            left: 0;

            background: #000000;

            color: #ffffff;

            padding: 8px 12px;

            z-index: 100;

            text-decoration: none;

            font-weight: bold;

        }

        .skip-link:focus {

            top: 0;

        }

        /* Required-field asterisk colour supplemented with "(required)" text */

        .required-star {

            color: #c00000;

            font-weight: bold;

        }

        /* Screen-reader-only helper class */

        .sr-only {

            position: absolute;

            width: 1px;

            height: 1px;

            padding: 0;

            margin: -1px;

            overflow: hidden;

            clip: rect(0, 0, 0, 0);

            white-space: nowrap;

            border: 0;

        }

    </style>


```

```

/* Form layout – labels wrap their controls for tight proximity */

.form-field {
    margin-bottom: 1rem;
}

.form-field label {
    display: block;
    margin-bottom: 0.25rem;
}

.form-field input[type="text"],
.form-field input[type="email"],
.form-field textarea {
    display: block;
    margin-top: 0.15rem;
}

/* Radio group */

.radio-group {
    margin-bottom: 0.5rem;
}

.radio-option {
    display: inline-block;
    margin-right: 1rem;
}

.radio-option label {
    display: inline;
    cursor: pointer;
}

/* Error styling – uses border + icon, not colour alone (WCAG 1.4.1) */

.error-message {
    color: #c00000;
    font-weight: bold;
    margin-top: 0.25rem;
    display: none;
}

.error-message::before {
    content: "\26A0 ";
}

input:invalid:not(:placeholder-shown),
textarea:invalid:not(:placeholder-shown) {
    border: 2px solid #c00000;
    outline: 2px solid #c00000;
}

/* Button styling – distinct via border, weight, and background (WCAG 1.4.1) */

input[type="submit"],

```

```

        input[type="reset"] {

            padding: 0.5rem 1rem;

            font-size: 1rem;

            cursor: pointer;

            border: 2px solid #333;

            border-radius: 4px;

            margin-right: 0.5rem;

        }

        input[type="submit"] {

            background-color: #005a00;

            color: #ffffff;

            font-weight: bold;

        }

        input[type="reset"] {

            background-color: #f0f0f0;

            color: #333333;

        }

        .hint-text {

            display: block;

            font-size: 0.9em;

            color: #555;

            margin-top: 0.15rem;

        }

    </style>

</head>

<body>

    <!-- Skip-to-content link placed as first focusable element (WCAG 2.4.1) -->

    <a href="#main-content" class="skip-link">Skip to main form content</a>

    <main id="main-content">

        <h1>WAI Form Example</h1>

        <p>Please complete all required form fields indicated with <span class="required-star" aria-hidden="true">*</span><span class="sr-only">(required)</span> below to submit your order.</p>

        <!-- Live region for form-level error announcements (WCAG 3.3.1, 3.3.3) -->

        <div id="form-errors" role="alert" aria-live="assertive"></div>

        <form method="post" action="http://mercury.swin.edu.au/it000000/formtest.php" novalidate>

            <fieldset>

                <legend>Contact Information</legend>

                <!-- Full Name - label wraps input for proximity (WCAG 1.3.1 Check 211) -->

                <div class="form-field">

```

```

        <label for="cname">Full Name <span class="required-star" aria-hidden="true">*</span><span
class="sr-only"> (required)</span>:

        <input type="text" name="cname" id="cname" required aria-required="true" aria-
describedby="cname-error" autocomplete="name" placeholder=" " />

        </label>

        <span class="error-message" id="cname-error" role="alert">Please enter your full name.</span>

    </div>

    <!-- E-mail – label wraps input for proximity -->

    <div class="form-field">

        <label for="cemail">E-mail Address <span class="required-star" aria-hidden="true">*</span><span
class="sr-only"> (required)</span>:

        <input type="email" name="cemail" id="cemail" required aria-required="true" aria-
describedby="cemail-error" autocomplete="email" placeholder=" " />

        </label>

        <span class="error-message" id="cemail-error" role="alert">Please enter a valid e-mail
address.</span>

    </div>

    <!-- Mailing Address – label wraps textarea for proximity (Check 96) -->

    <div class="form-field">

        <label for="caddr">Mailing Address <span class="required-star" aria-hidden="true">*</span><span
class="sr-only"> (required)</span>:

        <textarea name="caddr" id="caddr" rows="4" cols="40" required aria-required="true" aria-
describedby="caddr-error" autocomplete="street-address" placeholder=" "></textarea>

        </label>

        <span class="error-message" id="caddr-error" role="alert">Please enter your mailing
address.</span>

    </div>

</fieldset>

<fieldset>

    <legend>Credit Card Information</legend>

    <!-- Card type radios – each label wraps its radio for proximity (Check 125) -->

    <div class="form-field">

        <div class="radio-group" role="radiogroup" aria-labelledby="card-type-label" aria-
required="true">

            <span id="card-type-label"><strong>Card Type <span class="required-star" aria-
hidden="true">*</span><span class="sr-only"> (required)</span></strong></span>

            <div class="radio-option">

                <label for="cvisa"><input type="radio" name="card" id="cvisa" value="visa"
required aria-required="true" /> Visa</label>

            </div>

            <div class="radio-option">

                <label for="cmc"><input type="radio" name="card" id="cmc" value="mc" required
aria-required="true" /> MasterCard</label>

            </div>

```

```

        </div>

        <span class="error-message" id="card-error" role="alert">Please select a card type.</span>

    </div>

    <!-- Card Number – label wraps input for proximity -->

    <div class="form-field">

        <label for="cnumber">Card Number <span class="required-star" aria-hidden="true">*</span><span
class="sr-only"> (required)</span>:

            <input type="text" name="number" id="cnumber" required aria-required="true" aria-
describedby="cnumber-hint cnumber-error" autocomplete="cc-number" inputmode="numeric" pattern="[0-9\s\~]+> " placeholder=" " />

            </label>

            <span id="cnumber-hint" class="hint-text">Enter your credit card number (digits only).</span>

            <span class="error-message" id="cnumber-error" role="alert">Please enter a valid card
number.</span>

        </div>

    <!-- Card Expiry – label wraps input for proximity -->

    <div class="form-field">

        <label for="cexpiry">Card Expiry <span class="required-star" aria-hidden="true">*</span><span
class="sr-only"> (required)</span>:

            <input type="text" name="expiry" id="cexpiry" required aria-required="true" aria-
describedby="cexpiry-hint cexpiry-error" autocomplete="cc-exp" pattern="\d{2}/\d{2}" placeholder="MM/YY" />

            </label>

            <span id="cexpiry-hint" class="hint-text">Format: MM/YY</span>

            <span class="error-message" id="cexpiry-error" role="alert">Please enter a valid expiry date in
MM/YY format.</span>

        </div>

    </fieldset>

    <!-- Submit / Reset – wrapped in labels so label text is not empty (Check 188) -->

    <div class="form-field">

        <label for="csubmit"><span class="sr-only">Submit order form</span>

            <input type="submit" id="csubmit" value="Submit order" aria-label="Submit order form" />

        </label>

        <label for="creset"><span class="sr-only">Clear order form</span>

            <input type="reset" id="creset" value="Clear order" aria-label="Clear order form" />

        </label>

    </div>

</form>

</main>

<!--

Client-side validation script (WCAG 3.3.1, 3.3.3, 3.3.4).

- Keyboard accessible: all interactions use standard form events (submit, click).

- No flicker / flash: the script only shows/hides static text; no animation or colour cycling.

- Colour is not used alone: errors use text + icon (warning triangle) + border.

```

-->

<script>

```
(function () {

    var form = document.querySelector('form');

    var errorsDiv = document.getElementById('form-errors');


    // Validate on submit – shows descriptive per-field errors and a summary
    form.addEventListener('submit', function (e) {

        var errors = [];

        var firstInvalid = null;


        // Clear previous inline errors

        var errorSpans = form.querySelectorAll('.error-message');

        for (var i = 0; i < errorSpans.length; i++) {

            errorSpans[i].style.display = 'none';

        }


        // Required text / email / textarea fields

        var fields = [

            { id: 'cname',   errorId: 'cname-error',   msg: 'Full Name is required.' },
            { id: 'cemail',  errorId: 'cemail-error',  msg: 'A valid E-mail Address is required.' },
            { id: 'caddr',   errorId: 'caddr-error',   msg: 'Mailing Address is required.' },
            { id: 'cnumber', errorId: 'cnumber-error', msg: 'Card Number is required.' },
            { id: 'cexpiry', errorId: 'cexpiry-error', msg: 'Card Expiry is required (MM/YY).' }

        ];


        for (var j = 0; j < fields.length; j++) {

            var field = document.getElementById(fields[j].id);

            if (!field.value.trim()) {

                errors.push(fields[j].msg);

                document.getElementById(fields[j].errorId).style.display = 'block';

                if (!firstInvalid) firstInvalid = field;

            }

        }


        // Radio button group

        var cardSelected = form.querySelector('input[name="card"]:checked');

        if (!cardSelected) {

            errors.push('Please select a Card Type (Visa or MasterCard).');

            document.getElementById('card-error').style.display = 'block';

            if (!firstInvalid) firstInvalid = document.getElementById('cvisa');

        }

    });

});
```

```

        if (errors.length > 0) {
            e.preventDefault();

            errorsDiv.innerHTML = '<strong>Please correct the following errors:</strong>' +

                '<ul>' +

                    errors.map(function (err) { return '<li>' + err + '</li>'; }).join('') +

                    '</ul>';

            if (firstInvalid) firstInvalid.focus();

        } else {

            errorsDiv.innerHTML = '';

        }

    });

    // Confirm before reset – prevents accidental data loss (WCAG 3.3.4)
    document.getElementById('creset').addEventListener('click', function (e) {

        if (!confirm('Are you sure you want to clear all form data? This action cannot be undone.')) {

            e.preventDefault();

        }

    });

})();

</script>

<noscript>

    <p>This form requires JavaScript for enhanced validation and error messages. Please enable JavaScript in your browser
    settings.</p>

</noscript>

</body>

</html>

```